
29 Linear Programming

Many problems take the form of maximizing or minimizing an objective, given limited resources and competing constraints. If you can specify the objective as a linear function of certain variables, and if you can specify the constraints on resources as equalities or inequalities on those variables, then you have a *linear-programming problem*. Linear programs arise in a variety of practical applications. We begin by studying an application in electoral politics.

A political problem

Suppose that you are a politician trying to win an election. Your district has three different types of areas—urban, suburban, and rural. These areas have, respectively, 100,000, 200,000, and 50,000 registered voters. Although not all the registered voters actually go to the polls, you decide that to govern effectively, you would like at least half the registered voters in each of the three regions to vote for you. You are honorable and would never consider supporting policies you don't believe in. You realize, however, that certain issues may be more effective in winning votes in certain places. Your primary issues are preparing for a zombie apocalypse, equipping sharks with lasers, building highways for flying cars, and allowing dolphins to vote.

According to your campaign staff's research, you can estimate how many votes you win or lose from each population segment by spending \$1,000 on advertising on each issue. This information appears in the table of Figure 29.1. In this table, each entry indicates the number of

thousands of either urban, suburban, or rural voters who would be won over by spending \$1,000 on advertising in support of a particular issue. Negative entries denote votes that would be lost. Your task is to figure out the minimum amount of money that you need to spend in order to win 50,000 urban votes, 100,000 suburban votes, and 25,000 rural votes.

You could, by trial and error, devise a strategy that wins the required number of votes, but the strategy you come up with might not be the least expensive one. For example, you could devote \$20,000 of advertising to preparing for a zombie apocalypse, \$0 to equipping sharks with lasers, \$4,000 to building highways for flying cars, and \$9,000 to allowing dolphins to vote. In this case, you would win $(20 \cdot -2) + (0 \cdot 8) + (4 \cdot 0) + (9 \cdot 10) = 50$ thousand urban votes, $(20 \cdot 5) + (0 \cdot 2) + (4 \cdot 0) + (9 \cdot 0) = 100$ thousand suburban votes, and $(20 \cdot 3) + (0 \cdot -5) + (4 \cdot 10) + (9 \cdot -2) = 82$ thousand rural votes. You would win the exact number of votes desired in the urban and suburban areas and more than enough votes in the rural area. (In fact, according to your model, in the rural area you would receive more votes than there are voters.) In order to garner these votes, you would have paid for $20 + 0 + 4 + 9 = 33$ thousand dollars of advertising.

policy	urban	suburban	rural
zombie apocalypse	-2	5	3
sharks with lasers	8	2	-5
highways for flying cars	0	0	10
dolphins voting	10	0	-2

Figure 29.1 The effects of policies on voters. Each entry describes the number of thousands of urban, suburban, or rural voters who could be won over by spending \$1,000 on advertising support of a policy on a particular issue. Negative entries denote votes that would be lost.

It's natural to wonder whether this strategy is the best possible. That is, can you achieve your goals while spending less on advertising? Additional trial and error might help you to answer this question, but a better approach is to formulate (or *model*) this question mathematically.

The first step is to decide what decisions you have to make and to introduce variables that capture these decisions. Since you have four decisions, you introduce four *decision variables*:

- x_1 is the number of thousands of dollars spent on advertising on preparing for a zombie apocalypse,
- x_2 is the number of thousands of dollars spent on advertising on equipping sharks with lasers,
- x_3 is the number of thousands of dollars spent on advertising on building highways for flying cars, and
- x_4 is the number of thousands of dollars spent on advertising on allowing dolphins to vote.

You then think about *constraints*, which are limits, or restrictions, on the values that the decision variables can take. You can write the requirement that you win at least 50,000 urban votes as

$$-2x_1 + 8x_2 + 0x_3 + 10x_4 \geq 50 . \quad (29.1)$$

Similarly, you can write the requirements that you win at least 100,000 suburban votes and 25,000 rural votes as

$$5x_1 + 2x_2 + 0x_3 + 0x_4 \geq 100 \quad (29.2)$$

and

$$3x_1 - 5x_2 + 10x_3 - 2x_4 \geq 25 . \quad (29.3)$$

Any setting of the variables x_1 , x_2 , x_3 , x_4 that satisfies inequalities (29.1)–(29.3) yields a strategy that wins a sufficient number of each type of vote.

Finally, you think about your *objective*, which is the quantity that you wish to either minimize or maximize. In order to keep costs as small as possible, you would like to minimize the amount spent on advertising. That is, you want to minimize the expression

$$x_1 + x_2 + x_3 + x_4 . \quad (29.4)$$

Although negative advertising often occurs in political campaigns, there is no such thing as negative-cost advertising. Consequently, you require that

$$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, \text{ and } x_4 \geq 0 . \quad (29.5)$$